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FLUX AGENT AND SEMICONDUCTOR PACKAGE INCLUDING CURED PRODUCT THEREOF

Abstract

Provided is a flux agent according to embodiments of the inventive concept, wherein the flux agent includes a first compound including an ester group and an epoxy group, a fourth compound which is a chemical reactant of a second compound including an epoxy group and a third compound including an amine group and a carboxyl group, and a photoinitiator, wherein the content of the first compound is 50 wt % to 60 wt % based on 95 wt % of the total agent, the content of the fourth compound is 3 wt % to 33 wt % based on 100 wt % of the total agent, and the content of the photoinitiator is 2 wt % to 10 wt % based on 100 wt % of the total agent.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This U.S. non-provisional patent application claims priority under 35 U.S.C. § 119 of Korean Patent Application Nos. 10-2024-0033871, filed on Mar. 11, 2024, and 10-2025-0030023, filled on Mar. 7, 2025, the entire contents of which are hereby incorporated by reference.

BACKGROUND

[0002] The present disclosure herein relates to a flux agent and a semiconductor package including a cured product of the flux agent.

[0003] Electronic devices have become smaller and more multi-functional according to the rapid development of the electronics industry and the demands of users. Accordingly, there is also an increasing need for semiconductor devices used in electronic devices to be small and multi-functional.

SUMMARY

[0004] The present disclosure provides a flux agent for performing various functions.

[0005] The present disclosure also provides a method for manufacturing a semiconductor package with reduced difficulty of a manufacturing process.

[0006] A flux agent according to embodiments of the inventive concept comprises a first compound comprising an ester group and an epoxy group, a fourth compound which is a chemical reactant of a second compound comprising an epoxy group and a third compound comprising an amine group and a carboxyl group, and a photoinitiator, wherein the content of the first compound is 60 wt % to 95 wt % based on 100 wt % of the total agent, the content of the fourth compound is 3 wt % to 33 wt % based on 100 wt % of the total agent, and the content of the photoinitiator is 2 wt % to 10 wt % based on 100 wt % of the total agent.

[0007] A semiconductor package according to some embodiments of the inventive concept comprises a semiconductor substrate comprising an upper surface and a lower surface which are opposite to each other, a semiconductor chip comprising a first surface and a second surface which are opposite to each other, and disposed on the upper surface of the semiconductor substrate, wherein the first surface faces the upper surface of the semiconductor substrate, and a plurality of protective patterns interposed between the semiconductor substrate and the semiconductor chip, wherein each of the plurality of protective patterns is a cured product of a flux agent, wherein the flux agent comprises a first compound comprising an ester group and an epoxy group, a fourth compound which is a chemical reactant of a second compound comprising an epoxy group and a third compound comprising an amine group and a carboxyl group, and a photoinitiator, wherein the content of the first compound is 60 wt % to 95 wt % based on 100 wt % of the flux agent, the content of the fourth compound is 3 wt % to 33 wt % based on 100 wt % of the flux agent, and the content of the photoinitiator is 2 wt % to 10 wt % based on 100 wt % of the flux agent.

[0008] A semiconductor package according to some embodiments of the inventive concept comprises a semiconductor substrate comprising an upper surface and a lower surface which are opposite to each other, a plurality of first conductive pads on the upper surface of the semiconductor substrate, a semiconductor chip including a first surface and a second surface which

are opposite to each other, and disposed on the upper surface of the semiconductor substrate, wherein the first surface faces the upper surface of the semiconductor substrate, a plurality of second conductive pads on the first surface of the semiconductor chip, a plurality of connecting bumps interposed between the plurality of first conductive pads and the plurality of second conductive pads, and a plurality of protective patterns covering the plurality of connecting bumps, wherein a width in a first direction of each of the plurality of protective patterns decreases in a third direction farther from the upper surface of the semiconductor substrate, wherein the first direction is a direction parallel to the upper surface of the semiconductor substrate, wherein the third direction is a direction perpendicular to the upper surface of the semiconductor substrate.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0009] The accompanying drawings are included to provide a further understanding of the inventive concept, and are incorporated in and constitute a part of this specification. The drawings illustrate embodiments of the inventive concept and, together with the description, serve to explain principles of the inventive concept. In the drawings:

[0010] FIG. 1 is a plan view of a semiconductor package according to embodiments of the inventive concept;

[0011] FIG. 2 is a cross-sectional view taken along A-A' of FIG. 1;

[0012] FIG. 3 to FIG. 8 are views showing a method for manufacturing a semiconductor package according to embodiments of the inventive concept, and are cross-sectional views corresponding to A-A' of FIG. 1;

[0013] FIG. 9 to FIG. 14 are views showing a method for manufacturing a semiconductor package according to some embodiments of the inventive concept, and are cross-sectional views corresponding to A-A' of FIG. 1;

[0014] FIG. 15 is a cross-sectional view of a semiconductor package according to some embodiments of the inventive concept, and are cross-sectional views taken along A-A' of FIG. 1;

[0015] FIG. 16 to FIG. 19 are views showing a method for manufacturing a semiconductor package according to some embodiments of the inventive concept, and are cross-sectional views corresponding to A-A' of FIG. 1;

[0016] FIG. 20 is a cross-sectional view of a semiconductor package according to some embodiments of the inventive concept, and are cross-sectional views taken along A-A' of FIG. 1;

[0017] FIG. 21 to FIG. 25 are views showing a method for manufacturing a semiconductor package according to some embodiments of the inventive concept, and are cross-sectional views corresponding to A-A' of FIG. 1;

[0018] FIG. 26 is a cross-sectional view according to some embodiments of the inventive concept, and is a cross-sectional view taken along line A-A' of FIG. 1;

[0019] FIG. 27 is a plan view of a semiconductor package according to some embodiments of the inventive concept;

[0020] FIG. 28 is a cross-sectional view of a semiconductor package according to some embodiments of the inventive concept, and is a cross-sectional view taken along A-A' of FIG. 27;

[0021] FIG. 29 is a plan view of a semiconductor package according to some embodiments of the inventive concept;

[0022] FIG. 30 is a cross-sectional view of a semiconductor package according to some embodiments of the inventive concept, and is a cross-sectional view taken along A-A' of FIG. 29;

[0023] FIG. 31A and FIG. 31B are results of photographing a semiconductor chip and a semiconductor substrate using a scanning electron microscope;

[0024] FIG. 32 is result of photographing a protection pattern according to an embodiment formed

on a semiconductor chip using a scanning electron microscope; and
[0025] FIG. **33** is result of photographing a cross-section of a junction according to an embodiment using a scanning electron microscope;

DETAILED DESCRIPTION

[0026] Throughout the present specification, reference to “one embodiment” or “an embodiment” means that a particular feature, structure, or characteristic described in connection with the embodiment may be included in at least one embodiment disclosed herein. Therefore, in various places throughout the present specification, the appearances of phrases “in one embodiment,” “in an embodiment,” or “according to an embodiment” (or other phrases with similar meanings) do not necessarily all refer to the same embodiment. In addition, particular features, structures, or characteristics may be combined in any suitable manner in one or more embodiments. In this regard, as used herein, the word “exemplary” means “providing an example, exemplification, or illustration.” Any embodiment described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments. In addition, depending on the context of the discussion herein, a singular term may include a corresponding plural form, and a plural term may include a corresponding singular form. It should be noted that the various drawings (including block configuration diagram) shown and discussed herein are for illustrative purposes only, and are not drawn to scale.

[0027] In the present specification, unless otherwise defined, a particle diameter may be an average particle diameter. In addition, the particle diameter refers to an average particle diameter (D.sub.50) meaning a diameter of a particle having a cumulative volume of 50 vol % in a particle size distribution. The average particle diameter (D.sub.50) may be measured by a method widely known to those skilled in the art, for example, by using a particle size analyzer, or using a transmission electron microscope (TEM) photograph or scanning electron microscope (SEM) photograph. As another method, a measurement device using dynamic light-scattering may be used to perform measurement, the number of particles for each particle size range may be counted by performing data analysis, and then an average particle diameter (D.sub.50) value may be obtained by performing calculated therefrom. Alternatively, the average particle diameter (D.sub.50) may be measured by a laser diffraction method. More specifically, when the measurement is performed by a laser diffraction method, particles to be measured may be dispersed in a dispersion medium, introduced into a commercially available laser diffraction particle diameter measurement device (e.g., MT 3000 manufactured by Microtrac Co., Ltd.), and then irradiated with an ultrasonic wave of about 28 kHz to an output of 60 W to calculate an average particle diameter (D.sub.50) at 50% of a particle diameter distribution in the measurement device.

[0028] Hereinafter, embodiments of the inventive concept will be described with reference to the accompanying drawings to describe the inventive concept in detail.

[0029] FIG. **1** is a plan view of a semiconductor package according to embodiments of the inventive concept. FIG. **2** is a cross-sectional view taken along A-A' of FIG. **1**.

[0030] Referring to FIG. **1** and FIG. **2**, a semiconductor substrate **1000** including an upper surface **1000a** and a lower surface **1000b** which are opposite to each other may be provided. The semiconductor substrate **1000** may be, for example, a printed circuit board (PCB), a silicon substrate, glass substrate.

[0031] A plurality of first conductive pads **1010** may be disposed on the upper surface **1000a** of the semiconductor substrate **1000**. The plurality of first conductive pads **1010** may be spaced apart from each other in a first direction **D1** and a second direction **D2**.

[0032] In the present specification, the first direction **D1** may be a direction parallel to the upper surface **1000a** of the semiconductor substrate **1000**. The second direction **D2** may be a direction parallel to the upper surface **1000a** of the semiconductor substrate **1000** and intersecting the first direction **D1**. The third direction **D3** may be a direction perpendicular to the upper surface **1000a** of the semiconductor substrate **1000**.

[0033] The plurality of first conductive pads **1010** may include, for example, one or more selected from the group consisting of copper (Cu), aluminum (Al), tungsten (W), titanium (Ti) nickel (Ni), gold (Au), and an alloy thereof. The disposition and number of the plurality of first conductive pads **1010** may not be limited to that illustrated in FIG. 1. The plurality of first conductive pads **1010** may be fewer than nine, or may be provided more than nine.

[0034] A semiconductor chip **100** including a first surface **100a** and a second surface **100b** which are opposite to each other may be provided on the upper surface **1000a** of the semiconductor substrate **1000**. The first surface **100a** of the semiconductor chip **100** may face the upper surface **1000a** of the semiconductor substrate **1000**. The semiconductor chip **100** may include, for example, one or more among a passive element, an active element, a light emitting diode, a memory chip, and a logic chip. The semiconductor chip **100** may include, for example, a volatile memory semiconductor chip such as a dynamic random access memory (DRAM) or a static random access memory (SRAM).

[0035] The semiconductor chip **100** may include, for example, a non-volatile memory semiconductor chip such as a flash memory, a phase-change random access memory (PRAM), a magnetic random access memory (MRAM), a ferroelectric random access memory (FeRAM), or a resistive random access memory (RRAM).

[0036] A plurality of second conductive pads **110** may be disposed on the first surface **100a** of the semiconductor chip **100**. Each of the plurality of second conductive pads **110** may be disposed to correspond to each of the plurality of first conductive pads **1010**. The plurality of second conductive pads **110** may include, for example, one or more selected from the group consisting of copper (Cu), aluminum (Al), tungsten (W), titanium (Ti) nickel (Ni), gold (Au), and an alloy thereof.

[0037] A plurality of connecting bumps **210** may be interposed between the plurality of first conductive pads **1010** and the plurality of second conductive pads **110**.

[0038] Each of the plurality of connecting bumps **210** may be interposed between corresponding first and second conductive pads among the plurality of first and second conductive pads **1010** and **110**. Accordingly, the first and second conductive pads **1010** and **110** may be electrically connected to each other, and the semiconductor substrate **1000** and the semiconductor chip **100** may be electrically connected to each other. The plurality of connecting bumps **210** may be, for example, solder.

[0039] A plurality of protective patterns **500** may be interposed between the semiconductor chip **100** and the semiconductor substrate **1000**. The plurality of protective patterns **500** may come into contact with the upper surface **1000a** of the semiconductor substrate **1000** and the first surface **100a** of the semiconductor chip **100**. The plurality of protective patterns **500** may be spaced apart from each other in the first direction D1 and the second direction D2.

[0040] The plurality of protective patterns **500** may cover the plurality of first and second conductive pads **1010** and **110** and the connecting bumps **210**. More specifically, each of the plurality of protective patterns **500** may cover a side surface **1010S** of a corresponding first conductive pad and a side surface **110S** of a corresponding second conductive pad. Each of the plurality of protective patterns **500** may form, for example, a closed curve when viewed in a plan view. The plurality of protective patterns **500** may protect the plurality of first and second conductive pads **1011** and **110**, and the plurality of connecting bumps **210** from external impurities and impacts.

[0041] A width 500 W of each of the plurality of protective patterns **500** in the first direction D1 may be smaller than a width 100 W of the semiconductor chip **100** in the first direction D1. The width 500 W of each of the plurality of protective patterns **500** in the first direction D1 may decrease in the third direction D3 farther from the upper surface **1000a** of the semiconductor substrate **1000**. The width 500 W of each of the plurality of protective patterns **500** in the first direction D1 may increase closer to the upper surface **1000a** of the semiconductor substrate **1000**.

The width 500 W of each of the plurality of protective patterns **500** in the first direction **D1** may decrease closer to the first surface **100a** of the semiconductor chip **100**. A maximum width of each of the plurality of protective patterns **500** in the first direction **D1** may be larger than a width 110 W of each of the plurality of first conductive pads **1010** in the first direction **D1**.

[0042] The plurality of protective patterns **500** may be a cured product of a flux agent. The flux agent may include a first compound including an ester group and an epoxy group, a fourth compound, and a photoinitiator.

[0043] The content of the first compound may be 60 wt % to 95 wt % based on 100 wt % of the flux agent. If the content of the first compound satisfies the above-described numerical range, the formation of the protective pattern **500** on the semiconductor chip **100**, the semiconductor substrate **1000**, the plurality of first and second conductive pads, and connecting bumps may effectively performed and after curing the protective pattern **500** may allow sufficient fluidity during the bonding process.

[0044] The first compound may include at least one of glycidyl epoxy and cyclohexene oxide. The first compound may include, for example, one or more selected from the group consisting of Methyl 3-(7-oxabicyclo[4.1.0]heptan-3-yl)-7-oxabicyclo[4.1.0]heptane-3-carboxylate, (5-Methyl-7-oxabicyclo[4.1.0]heptan-3-yl)methyl 5-methyl-7-oxabicyclo[4.1.0]heptane-3-carboxylate, [2,2-Dimethyl-3-(6-methyl-7-oxabicyclo[4.1.0]heptane-3-carbonyl)oxypropyl] 6-methyl-7-oxabicyclo[4.1.0]heptane-3-carboxylate, (1-Methyl-7-oxabicyclo[4.1.0]heptan-3-yl)methyl 1-methyl-7-oxabicyclo[4.1.0]heptane-3-carboxylate, 7-Oxabicyclo[4.1.0]heptan-2-ylmethyl 7-oxabicyclo[4.1.0]heptane-3-carboxylate, 6-(7-Oxabicyclo[4.1.0]heptan-1-ylmethyl)-7-oxabicyclo[4.1.0]heptane-3-carboxylate, (2-Methyl-7-oxabicyclo[4.1.0]heptan-3-yl)methyl 2-methyl-7-oxabicyclo[4.1.0]heptane-3-carboxylate, 2-(7-Oxabicyclo[4.1.0]heptan-3-yl) ethyl 7-oxabicyclo[4.1.0]heptane-3-carboxylate, 2-[2-(7-Oxabicyclo[4.1.0]heptane-3-carbonyloxy)ethoxy]ethyl 7-oxabicyclo[4.1.0]heptane-3-carboxylate, Bis(7-oxabicyclo[4.1.0]heptan-3-ylmethyl) heptanedioate, 3,4-Epoxy cyclohexylmethyl 3,4-epoxycyclohexanecarboxylate, Bis(3,4-epoxycyclohexylmethyl) adipate, Chissonox 3-[2-(3-Carboxylato-7-oxabicyclo[4.1.0]heptan-3-yl)ethyl]-7-201, oxabicyclo[4.1.0]heptane-3-carboxylate, 2-(7-Oxabicyclo[4.1.0]heptane-3-carbonyloxy) ethyl 7-oxabicyclo[4.1.0]heptane-3-carboxylate, 3,4-Epoxy cyclohexyl-3,4-epoxycyclo-hexane carboxylate, 5-Methyl-3-[(5-methyl-7-oxabicyclo[4.1.0]heptan-3-yl)methyl]-7-oxabicyclo[4.1.0]heptane-3-carboxylate, 1,4-Cyclohexanedimethanol bis(3,4-epoxycyclohexanecarboxylate), Cyclohexylmethyl-3,4-epoxycyclohexanecarboxylate, Glycidyl methacrylate, Ethyl 2,3-epoxybutyrate, Methyl 3,4-epoxybutyrate, Ethyl oxirane-2-carboxylate, Glycidyl butyrate, (2-Methyloxiranyl)methyl methacrylate, 9,10-Epoxy stearate, Diglycidyl isophthalate, oxiran-2-ylmethyl benzoate, Tris(oxiranylmethyl)benzene-1,3,5-tricarboxylate, Bis(2,3-epoxypropyl) terephthalate, Bis(oxiran-2-ylmethyl) 5-carbonochloridoylbenzene-1,3-dicarboxylate, and a combination thereof.

[0045] The content of fourth compound may be 3 wt % to 33 wt % based on 100 wt % of the flux agent. The fourth compound may be a chemical reactant of a second compound including an epoxy group and a third compound including an amine group and a carboxyl group.

[0046] If the content of the fourth compound satisfies the above-described numerical range, a metal oxide on the semiconductor chip **100**, the semiconductor substrate **1000**, and the plurality of first and second conductive pads may be effectively removed, and the formation of the protective pattern may be effectively performed. For example, if the content of the fourth compound is less than 3 wt % a metal oxide may not be removed effectively, the content of the fourth compound is greater than 33 wt % the protective pattern may not be formed effectively.

[0047] The fourth compound may include a covalent bond formed by the reaction between the epoxy group of the second compound and the amine group of the third compound. The fourth compound may include a covalent bond formed by the reaction between the epoxy group of the second compound and the carboxyl group of the third compound. The fourth compound may not be

a simple mixture of the second compound and the third compound, and may be a chemical reactant of the second compound and the third compound. As an example, the fourth compound may be easily dissolved in an organic solvent.

[0048] The second compound may include, for example, at least one of glycidyl epoxy and cyclohexene oxide. The second compound may comprise, for example, an epoxy siloxane resin. The second compound may comprise, for example, one or more selected from the group consisting of 3-Glycidoxypropyltrimethoxysilane, 3-Glycidoxypropyltriethoxysilane, (3-Glycidoxypropyl)methyldiethoxysilane, (3-Glycidoxypropyl)methyldimethoxysilane, β -(3,4-Epoxycyclohexyl)ethyltrimethoxysilane, β -(3,4-Epoxycyclohexyl)ethyltriethoxysilane, 3,4-Epoxycyclohexylmethyl 3,4-epoxycyclohexanecarboxylate, Bis(3,4-epoxycyclohexylmethyl) adipate, and a combination thereof.

[0049] The third compound may include one or more selected from the group consisting of alpha (a)-amino acid, beta (B)-amino acid, gamma (γ)-amino acid, delta (δ)-amino acid, anthranilic acid, 3-aminobenzoic acid, para-aminobenzoic acid, and a combination thereof. The third compound may include, for example, may include one or more selected from the group consisting of selected from glycine, alanine, valine, leucine, isoleucine, lysine, arginine, histidine, aspartic acid, asparagine, glutamine, glutamic acid, phenylalanine, tyrosine, tryptophan, cysteine, methionine, serine, ornithine, 3-phenylserine, threonine, L-dopa, norleucine, penicillamine, sarcosine, proline, hydroxyproline, 3-hydroxyproline, 3,4-dihydroproline, pipercolic acid, β -alanine, 3-aminobutyric acid, isoserine, 3-aminoisobutyric acid, 3-amino-2-phenylpropionic acid, 3-amino-5-methylhexanoic acid, 3-amino-4-phenylbutyric acid, 3-amino-4-hydroxybutyric acid, 3-amino-4-hydroxypentanoic acid, 3-amino-4-methylpentanoic acid, 3-amino-3-phenylpropionic acid, pyrrolidine-3-carboxylic acid, γ -aminobutyric acid, 4-amino-3-hydroxybutyric acid, 3-pyrrolidine-2-yl-propionic acid, 3-aminocyclohexanecarboxylic acid, 4-guanidinobutyric acid, 4-aminobenzoic acid, 3-aminobenzoic acid, 2-aminobenzoic acid, 3,5-diaminobenzoic acid, 4-aminosalicylic acid, 5-aminosalicylic acid, 3-aminoisonicotinic acid, 4-aminonicotinic acid, 5-aminonicotinic acid, 2-aminonicotinic acid, 6-aminonicotinic acid, 2-aminoisonicotinic acid, 6-aminopicolinic acid, a combination thereof, and a chemical reactant thereof.

[0050] The content of the photoinitiator may be 2 wt % to 10 wt % based on 100 wt % of the flux agent. If the content of the photoinitiator is less than 2 wt %, a photo-curing reaction may not effectively occur. If the content of the photoinitiator is greater than 10 wt %, a metal oxide on the plurality of first and second conductive pads **1010** and **110** and the plurality of connecting bumps **210** may not be effectively removed.

[0051] The photoinitiator may initiate a curing reaction of the first compound including an ester group of the first compound by light energy or heat energy. The photoinitiator may form, for example, cations by light energy or heat energy, and may initiate the curing reaction through the cations.

[0052] The photoinitiator may include, for example, at least one selected from the group consisting of an onium salt (e.g., an iodonium salt, a sulfonium salt, a phosphonium salt, a diazonium salt, a pyridinium salt, or an imide) such as 3-methyl-2-butenyltetramethylenesulfonium hexafluoroantimonate salt, ytterbium trifluoromethanesulfonate salt, samarium trifluoromethanesulfonate salt, erbium trifluoromethanesulfonate salt, triarylsulfonium hexafluoroantimonate salt, triarylsulfonium hexafluorophosphate salt, lanthanum trifluoromethanesulfonate salt, tetrabutylphosphonium methanesulfonate salt, ethyltriphenylphosphonium bromide salt, diphenyldiodonium hexafluoroantimonate salt, diphenyldiodonium hexafluorophosphate salt, dithoriliodonium hexafluorophosphate salt, 9-(4-hydroxyethoxyphenyl)cyanthreneium hexafluorophosphate salt, and 1-(3-methylbut-2-enyl) tetrahydro-1H-thiophenium hexafluoroantimonate salt, 2-(9-oxoxanthen-2-yl) propionic acid 1,5,7-triazabicyclo[4.4.0]dec-5-ene salt, and a combination thereof.

[0053] The flux agent may further include a catalyst, a solvent, a leveling agent, and metal

nanoparticles. The catalyst may include, for example, one or more selected from the group consisting of a base such as 1,5,7-triazabicyclo[4.4.0]dec-5-ene, 2-methylimidazole, and 1-methylimidazole, a Lewis acid such as zinc acetate, a phosphine such as triphenylphosphine, dibutyl phosphate, and a combination thereof.

[0054] The solvent may be, for example, an organic solvent, for example, one or more among acetone, benzene, toluene, hexane, methanol, dichloromethane, ethyl acetate, γ -Butyrolactone, Cyclopentanone, Propylene Glycol Monomethyl Ether, Propylene Glycol Monomethyl Ether Acetate and diethyl ether.

[0055] The metal nanoparticles may contain one or more among a metal such as tin (Sn), copper (Cu), silver (Ag), bismuth (Bi), indium (In), lead (Pd), cadmium (Cd), antimony (Sb), gallium (Ga), arsenic (As), germanium (Ge), zinc (Zn), aluminum (Al), gold (Au), silicon (Si), nickel (Ni), and phosphorus (P), a non-metal, and an alloy selected from a combination thereof. The average particle diameter of the metal nanoparticles may be, for example, 0.01 μm to 1 μm .

[0056] A molding film **400** may be disposed on the upper surface **1000a** of the semiconductor substrate **1000**. The molding film **400** may cover the semiconductor chip **100**, and may fill a space between the plurality of protective patterns **500**. The molding film **400** may come into contact with the upper surface **1000a** of the semiconductor substrate **1000**, and may come into contact with the first surface **100a** of the semiconductor chip **100**. The molding film **400** may come into contact with a side surface **500S** of the plurality of protective patterns **500**. The molding film **400** may include, for example, an epoxy molding compound, molded underfill. The molding film **400** may protect the semiconductor package from impurities, and external impacts.

[0057] FIG. **3** to FIG. **8** are views showing a method for manufacturing a semiconductor package according to embodiments of the inventive concept, and are cross-sectional views corresponding to A-A' of FIG. **1**. In order to simplify the description, the same descriptions as those of the semiconductor package described with reference to FIG. **1** and FIG. **2** will be omitted.

[0058] Referring to FIG. **3**, a flux agent **510P** may be prepared. The preparing of the flux agent **510P** may include, for example, mixing a first compound, a fourth compound, and a photoinitiator in a solvent. As an example, the fourth compound may be easily dissolved in an organic solvent. The content of the first compound may be 60 wt % to 95 wt % based on 100 wt % of the flux agent. The first compound may include at least one of glycidyl epoxy and cyclohexene oxide.

[0059] The content of fourth compound may be 3 wt % to 33 wt % based on 100 wt % of the flux agent. The fourth compound may be a chemical reactant of a second compound including an epoxy group and a third compound including an amine group and a carboxyl group. The second compound may include, for example, at least one of glycidyl epoxy and cyclohexene oxide. The second compound may include, for example, an epoxy siloxane resin. The forming of the fourth compound may include, for example, mixing the second compound and the third compound at room temperature, and reacting the second compound and the third compound at 150° C. to 300° C. for 30 minutes to 24 hours. The content of the photoinitiator may be 2 wt % to 10 wt % based on 100 wt % of the flux agent. The flux agent may further include a catalyst, a solvent, a leveling agent, and metal nanoparticles.

[0060] A plurality of first conductive pads **1010** may be formed on an upper surface **1000a** of a semiconductor substrate **1000**. The flux agent **510P** may be formed on the upper surface **1000a** of the semiconductor substrate **1000**. The flux agent **510P** may cover the upper surface **1000a** of the semiconductor substrate **1000**, and may cover a plurality of first conductive pads **1010** on the semiconductor substrate **1000**. The flux agent **510P** may conformally cover side surfaces and upper surfaces of the plurality of first conductive pads **1010**. The forming of the flux agent **510P** on the upper surface **1000a** of the semiconductor substrate **1000** may include, for example, a spin coating process.

[0061] Referring to FIG. **4**, a plurality of preliminary protective patterns **500P** may be formed. The plurality of preliminary protective patterns **500P** may be formed on the plurality of first conductive

pads **1010**. The forming of the plurality of preliminary protective patterns **500P** may include, for example, forming a first mask pattern **MP1** on the flux agent **510P**, wherein the first mask pattern **MP1** exposes an upper surface **510P_U** of the flux agent **510P** corresponding to an upper surface **1010U** of each of the plurality of first conductive pads **1010**, and curing the flux agent by irradiating the exposed upper surface **510P_U** of the flux agent **510P** with light or heat energy **610**. [0062] By the light energy or heat energy, a curing reaction by the photoinitiator may occur. A curing reaction of the first compound of the flux agent **510P** may occur. Accordingly, the flux agent **510P** may be cured. The light energy may include, for example, light (e.g., ultraviolet light) having a wavelength of 10 nm to 600 nm.

[0063] According to the embodiments of the inventive concept, the plurality of preliminary protective patterns **500P**, which are cured products of the flux agent **510P**, may be formed on the plurality of first conductive pads **1010**. Each of the plurality of preliminary protective patterns **500P** may be in a solid state at room temperature.

[0064] Referring to FIG. 5, the first mask pattern **MP1** may be removed. An upper surface **500P_U** of the plurality of preliminary protective patterns **500P** and the upper surface **510P_U** of the flux agent **510P** may be exposed.

[0065] The plurality of preliminary protective patterns **500P** may be disposed on the plurality of first conductive pads **1010**. The flux agent **510P** may cover a side surface of each of the plurality of preliminary protective patterns **500P**, and a side surface of each of the plurality of first conductive pads **1010**.

[0066] Referring to FIG. 6, the flux agent **510P** may be removed. The plurality of preliminary protective patterns **500P** are in a solid state, and may not be removed. The upper surface **1000a** of the semiconductor substrate **1000** may be exposed, and the side surfaces of the plurality of first conductive pads **1010** and the plurality of preliminary protective patterns **500P** may be exposed. Furthermore, a defect in which the plurality of preliminary protective patterns **500P** flow or evaporate during a process may not occur. The semiconductor substrate **1000** on which the plurality of preliminary protective patterns **500P** are formed may be stored for a long period of time. Application fixation of the flux agent **510P** and a bonding process to be described later may be spatially and temporally dualized. Accordingly, a method for manufacturing various semiconductor packages may be applied. In addition, prior to a bonding process of the semiconductor substrate **1000** and a semiconductor chip **100**, the plurality of first conductive pads **1010** may be physically and chemically protected by the plurality of preliminary protective patterns **500P**.

[0067] Referring to FIG. 7, a plurality of second conductive pads **110** and a plurality of connecting bumps **210** may be formed on a first surface **100a** of the semiconductor chip **100**. Each of the plurality of connecting bumps **210** may be formed on a corresponding second conductive pad among the plurality of second conductive pads **110**.

[0068] The first surface **100a** of the semiconductor chip **100** may be provided to face the upper surface **1000a** of the semiconductor substrate **1000**. Each of the plurality of second conductive pads **110** and the plurality of connecting bumps **210** may be provided to correspond to a corresponding first conductive pad **1010** and a corresponding preliminary protective pattern **500P**.

[0069] Referring to FIG. 8, a bonding process **620** may be performed. The bonding process **620** may include applying pressure to the semiconductor chip **100** and the semiconductor substrate **1000** which are aligned. The bonding process **620** may include applying pressure to the semiconductor chip **100** and the semiconductor substrate **1000**, and applying light energy or heat energy to the plurality of preliminary protective patterns **500P**.

[0070] Referring back to FIG. 1 and FIG. 2, the semiconductor chip **100** and the semiconductor substrate **1000** may be bonded to each other by the bonding process. The plurality of first conductive pads **1010**, the plurality of second conductive pads **110**, and the plurality of connecting bumps **210** may be electrically connected to each other. Accordingly, the semiconductor chip **100** and the semiconductor substrate **1000** may be electrically connected to each other.

[0071] A plurality of protective patterns **500** may be formed. The forming of the plurality of protective patterns **500** may include, through the above-described bonding process, applying pressure to the plurality of preliminary protective patterns **500P**, and applying heat or light energy to the plurality of preliminary protective patterns **500P**. The light energy may comprise, for example, a laser having a wavelength of 600 nm to 3000 nm and microwave having a wavelength of 1 mm to 1 m.

[0072] Due to the heat energy or light energy in the bonding process, the plurality of preliminary protective patterns **500P** may have fluidity. The plurality of preliminary protective patterns **500P** may be pushed to side surfaces of the plurality of first and second conductive pads **1010** and **110** and the plurality of connecting bumps **210**. Due the above-described fluidity, a width **500 W** of each of the plurality of preliminary protective patterns **500P** in a first direction **D1** may decrease in a third direction **D3** farther from the upper surface **1000a** of the semiconductor substrate **1000**.

[0073] Furthermore, due to the heat energy or light energy, the fourth compound in the plurality of preliminary protective patterns **500P** may act as a reducing agent. Accordingly, a metal oxide on the plurality of connecting bumps **210** may be removed, and defects due to impurities during the bonding process may be removed.

[0074] The plurality of protective patterns **500** may cover some or all of side surfaces **1010S**, **110S**, and **500S** of the plurality of first and second conductive pads **1010** and **110** and the plurality of connecting bumps **210**, and may protect the plurality of first and second conductive pads **1010** and **110** and the plurality of connecting bumps **210**.

[0075] According to the embodiments of the inventive concept, the plurality of preliminary protective patterns **500P** may function as a flux agent for removing the metal oxide on the plurality of connecting bumps, and serve to protect the plurality of connecting bumps during a semiconductor manufacturing process.

[0076] A molding film **400** may be formed on the upper surface **1000a** of the semiconductor substrate **1000**. The molding film **400** may fill a space between the upper surface **1000a** of the semiconductor substrate **1000**, the semiconductor chip **100**, and the plurality of protective patterns **500**.

[0077] FIG. **9** to FIG. **14** are views showing a method for manufacturing a semiconductor package according to some embodiments of the inventive concept, and are cross-sectional views corresponding to A-A' of FIG. **1**. In order to simplify the description, the same descriptions as those of the method for manufacturing a semiconductor package described with reference to FIG. **1** and FIG. **8** will be omitted.

[0078] Referring to FIG. **9**, a flux agent **510P** may be applied on a first surface **100a** of a semiconductor chip **100**. The flux agent **510P** may cover the first surface **100a** of the semiconductor chip **100**, and may conformally cover a plurality of second conductive pads **110** and a plurality of connecting bumps **210** on the first surface **100a** of the semiconductor chip **100**. The applying of the flux agent **510P** on the first surface **100a** of the semiconductor chip **100** may include, for example, a spin coating process.

[0079] Referring to FIG. **10**, a plurality of preliminary protective patterns **500P** may be formed. An upper surface **500P_U** of the plurality of preliminary protective patterns **500P** may be positioned at a higher level than an upper surface **210U** of a plurality of connecting bumps **210**. The plurality of preliminary protective patterns **500P** may cover the plurality of second conductive pads **110** and the plurality of connecting bumps **210**. The plurality of preliminary protective patterns **500P** may cover side surfaces of the plurality of second conductive pads **110** and the plurality of connecting bumps **210**.

[0080] The forming of the plurality of preliminary protective patterns **500P** may include forming a second mask pattern **MP2** on the flux agent **510P**, wherein the second mask pattern **MP2** exposes an upper surface **510P_U** of the flux agent **510P** corresponding to an upper surface **110U** of each of the plurality of second conductive pads **110**, and curing the flux agent **510P** by irradiating the

exposed upper surface **510P_U** of the flux agent **510P** with light energy or heat energy **610**.

[0081] By the light energy or heat energy, a curing reaction by a photoinitiator may occur. A curing reaction of a first compound of the flux agent **510P** may occur. The light energy may include, for example, light (e.g., ultraviolet light) having a wavelength of 10 nm to 600 nm.

[0082] According to some embodiments of the inventive concept, the plurality of preliminary protective patterns **500P**, which are cured products of the flux agent **510P**, may be formed on the plurality of second conductive pads **110** and the plurality of connecting bumps **210**. Each of the plurality of preliminary protective patterns **500P** may be in a solid state at room temperature.

[0083] Referring to FIG. **11**, the second mask pattern **MP2** may be removed. An upper surface **500P_U** of the plurality of preliminary protective patterns **500P** and the upper surface **510P_U** of the flux agent **510P** may be exposed.

[0084] Referring to FIG. **12**, the flux agent **510P** may be removed. Accordingly, the first surface **100a** of the semiconductor chip **100** may be exposed, and the side surfaces of the plurality of second conductive pads **110** and the plurality of preliminary protective patterns **500P** may be exposed. The plurality of preliminary protective patterns **500P** are in a solid state, and may not be removed. Also, a defect in which the plurality of preliminary protective patterns **500P** flow or evaporate during a process of manufacturing a semiconductor package may not occur.

[0085] Furthermore, the semiconductor chip **100** on which the plurality of preliminary protective patterns **500P** are formed may be stored for a long period of time. Application fixation of the flux agent **510P** and a bonding process to be described later may be spatially and temporally dualized. A method for manufacturing various semiconductor packages may be applied.

[0086] In addition, prior to a bonding process of the semiconductor substrate **1000** and a semiconductor chip **100**, the plurality of second conductive pads **210** and the plurality of connecting bumps **210** may be physically and chemically protected by the plurality of preliminary protective patterns **500P**.

[0087] Referring to FIG. **13**, a plurality of first conductive pads **1010** may be formed on an upper surface **1000a** of a semiconductor substrate **1000**. The first surface **100a** of the semiconductor chip **100** may be provided to face the upper surface **1000a** of the semiconductor substrate **1000**. Each of the plurality of second conductive pads **110** and the plurality of connecting bumps **210** may be aligned on a corresponding first conductive pad among the plurality of first conductive pads **1010**.

[0088] Referring to FIG. **14**, a bonding process **620** may be performed. The bonding process **620** may include applying pressure to the semiconductor chip **100** and the semiconductor substrate **1000** which are aligned. The bonding process **620** may include applying pressure to the semiconductor chip **100** and the semiconductor substrate **1000**, and applying light energy or heat energy to the plurality of preliminary protective patterns **500P**.

[0089] The rest of the manufacturing method may be substantially the same as the manufacturing method of the semiconductor package described with reference to FIG. **1** and FIG. **8**.

[0090] FIG. **15** is a cross-sectional view of a semiconductor package according to some embodiments of the inventive concept, and are cross-sectional views taken along A-A' of FIG. **1**. In order to simplify the description, the same descriptions as those of the semiconductor package described with reference to FIG. **1** and FIG. **2** will be omitted.

[0091] Referring to FIG. **1** and FIG. **15**, a plurality of protective patterns **500** may be interposed between a semiconductor chip **100** and a semiconductor substrate **1000**. The plurality of protective patterns **500** may come into contact with an upper surface **1000a** of the semiconductor substrate **1000** and a first surface **100a** of the semiconductor chip **100**. The plurality of protective patterns **500** may be spaced apart from each other in a first direction **D1** and a second direction **D2**.

[0092] The plurality of protective patterns **500** may cover a plurality of first and second conductive pads **1010** and **110** and a plurality of connecting bumps **210**. More specifically, each of the plurality of protective patterns **500** may cover a side surface **1010S** of a corresponding first conductive pad **1010** and a side surface **110S** of a corresponding second conductive pad **110**. Each of the plurality

of protective patterns **500** may conformally cover corresponding first and second conductive pads **1010** and **110** and a corresponding connecting bump **210**.

[0093] Each of the plurality of protective patterns **500** may form, for example, a closed curve when viewed in a plan view. The plurality of protective patterns **500** may protect the plurality of first and second conductive pads **1011** and **110**, and the plurality of connecting bumps **210** from external impurities and impacts.

[0094] A width **500 W** of each of the plurality of protective patterns **500** in the first direction **D1** may be smaller than a width **100 W** of the semiconductor chip **100** in the first direction **D1**. The width **500 W** of each of the plurality of protective patterns **500** in the first direction **D1** may be larger than a width **1010 W** of each of the plurality of first conductive pads **1010** in the first direction **D1**. The width **500 W** of each of the plurality of protective patterns **500** in the first direction **D1** may be larger than a width **110 W** of each of the plurality of second conductive pads **110** in the first direction **D1**.

[0095] The rest of the components may be substantially the same as those of the semiconductor package described with reference to FIG. **1** and FIG. **2**.

[0096] FIG. **16** to FIG. **19** are views showing a method for manufacturing a semiconductor package according to some embodiments of the inventive concept, and are cross-sectional views corresponding to A-A' of FIG. **1**. In order to simplify the description, the same descriptions as those of the method for manufacturing a semiconductor package described with reference to FIG. **1** and FIG. **8** will be omitted.

[0097] Referring to FIG. **16**, a flux agent **510P** may be prepared in a storage unit **10**. The preparing of the flux agent **510P** may include, for example, preparing a first compound, a fourth compound, a photoinitiator, and a solvent in the storage unit **10**, and mixing the first compound, the fourth compound, the photoinitiator, and the solvent.

[0098] On the first surface **100a** of the semiconductor chip **100**, the plurality of second conductive pads **110** and the plurality of connecting bumps **210** on the plurality of second conductive pads **110** may be formed. On the plurality of second conductive pads **110** and the plurality of connecting bumps **210** on the first surface **100a** of the semiconductor chip **100**, the flux agent **510P** may be formed.

[0099] The forming of the flux agent **510P** on the first surface **100a** of the semiconductor chip **100** may include, for example, impregnating the plurality of second conductive pads **110** and the plurality of connecting bumps **210** on the first surface **100a** of the semiconductor chip **100** into the storage unit **10**.

[0100] Referring to FIG. **17**, a plurality of preliminary protective patterns **500P** may be formed on the first surface **100a** of the semiconductor chip **100**. The plurality of preliminary protective patterns **500P** may come into contact with the first surface **100a** of the semiconductor chip **100**, and may cover side surfaces of the plurality of second conductive pads **110**. The plurality of preliminary protective patterns **500P** may cover the plurality of connecting bumps **210**.

[0101] The forming of the plurality of preliminary protective patterns **500P** may include, for example, applying heat or light energy **610** to a flux agent on the first surface **100a** of the semiconductor chip **100**, thereby curing the flux agent. The light energy may include, for example, ultraviolet light having a wavelength of 10 nm to 600 nm.

[0102] Referring to FIG. **18**, a plurality of first conductive pads **1010** may be formed on an upper surface **1000a** of a semiconductor substrate **1000**. The first surface **100a** of the semiconductor chip **100** may be provided to face the upper surface **1000a** of the semiconductor substrate **1000**. Each of the plurality of second conductive pads **110** and the plurality of connecting bumps **210** may be alignment to correspond to a corresponding first conductive pad **1010**.

[0103] Referring to FIG. **19**, a bonding process **620** may be performed. The bonding process **620** may include applying pressure to the semiconductor chip **100** and the semiconductor substrate **1000** which are aligned. The bonding process **620** may include applying pressure to the

semiconductor chip **100** and the semiconductor substrate **1000**, and applying light energy or heat energy to the plurality of preliminary protective patterns **500P**.

[0104] According to some embodiments of the inventive concept, the plurality of preliminary protective patterns **500P**, which are cured products of the flux agent, may be formed on the plurality of second conductive pads **110** and the plurality of connecting bumps **210**. Each of the plurality of preliminary protective patterns **500P** may be in a solid state at room temperature. Therefore, a defect in which the preliminary protective patterns **500P** flow or evaporate during a process may not occur.

[0105] In addition, even if the first surface **100a** of the semiconductor chip **100** is disposed to face the upper surface **1000a** of the semiconductor substrate **1000**, the plurality of preliminary protective patterns **500p** may not flow. Therefore, the difficulty of a manufacturing process of a semiconductor package may be reduced. Furthermore, the semiconductor substrate **1000** on which the plurality

[0106] of preliminary protective patterns **500P** are formed may be stored for a long period of time. Accordingly, application fixation of the flux agent and a bonding process to be described later may be spatially and temporally dualized. Accordingly, a method for manufacturing various semiconductor packages may be applied.

[0107] Referring back to FIG. **1** and FIG. **15**, the plurality of protective patterns **500** may be formed. The forming of the plurality of protective patterns **500** may include, through the above-described bonding process, applying pressure to the plurality of preliminary protective patterns **500P**, and applying heat or light energy to the plurality of preliminary protective patterns **500P**. The light energy may include, for example, a laser having a wavelength of 600 nm to 3000 nm.

[0108] Due to the heat energy or light energy in the bonding process, the plurality of preliminary protective patterns **500P** may have fluidity.

[0109] The plurality of preliminary protective patterns **500P** may be pushed to side surfaces of the plurality of first and second conductive pads **1010** and **110** and the plurality of connecting bumps **210**. The plurality of protective patterns **500** may cover side surfaces **1010S**, **110S**, and **500S** of the plurality of first and second conductive pads **1010** and **110** and the plurality of connecting bumps **210**, and may protect the plurality of first and second conductive pads **1010** and **110** and the plurality of connecting bumps **210**.

[0110] Furthermore, due to the heat energy or light energy, a fourth compound in the plurality of preliminary protective patterns **500P** may act as a reducing agent. Accordingly, a metal oxide on the plurality of connecting bumps **210** may be removed, and defects due to impurities during the bonding process may be removed.

[0111] The rest of the manufacturing method may be substantially the same as the manufacturing method of the semiconductor package described with reference to FIG. **1** and FIG. **8**.

[0112] FIG. **20** is a cross-sectional view of a semiconductor package according to some embodiments of the inventive concept, and are cross-sectional views taken along A-A' of FIG. **1**. In order to simplify the description, the same descriptions as those of the semiconductor package described with reference to FIG. **1** and FIG. **2** will be omitted.

[0113] A protective layer **500L** may be disposed on a first surface **100a** of a semiconductor chip **100**. The protective layer **500L** may come into contact with the first surface **100a** of the semiconductor chip **100**, and may cover a side surface **110S** of each of a plurality of second conductive pads **110**.

[0114] Each of the plurality of protective patterns **500** may extend from the protective layer **500L** toward an upper surface **1000a** of a semiconductor substrate **1000**. The protective layer **500L** and the plurality of protective patterns **500** may be integrated with each other. The plurality of protective patterns **500** may be disposed between the semiconductor chip **100** and the semiconductor substrate **1000**. The plurality of protective patterns **500** may come into contact with the upper surface **1000a** of the semiconductor substrate **1000** and the first surface **100a** of the

semiconductor chip **100**. The plurality of protective patterns **500** may be spaced apart from each other in a first direction **D1** and a second direction **D2**.

[0115] The plurality of protective patterns **500** may cover a plurality of first and second conductive pads **1010** and **110** and a plurality of connecting bumps **210**. More specifically, each of the plurality of protective patterns **500** may cover a side surface **1010S** of a corresponding first conductive pad and a side surface **110S** of a corresponding second conductive pad. The protective layer **500L** and the plurality of protective patterns **500** may protect the plurality of first and second conductive pads, and the plurality of connecting bumps **210** from external impurities and impacts.

[0116] The protective layer **500L** and the plurality of protective patterns **500** may be cured products of a flux agent. The flux agent may include a first compound including an ester group and an epoxy group, a fourth compound, and a photoinitiator. The content of the first compound may be 60 wt % to 95 wt % based on 100 wt % of the flux agent. If the content of the first compound satisfies the above-described numerical range, the formation of the protective pattern on the semiconductor chip **100**, the semiconductor substrate **1000**, the plurality of first and second conductive pads, and connecting bumps may effectively performed and after curing the protective pattern may allow sufficient fluidity during the bonding process

[0117] The first compound may include at least one of glycidyl epoxy and cyclohexene oxide. The content of fourth compound may be 3 wt % to 33 wt % based on 100 wt % of the flux agent. The fourth compound may be a chemical reactant of a second compound including an epoxy group and a third compound including an amine group and a carboxyl group. The fourth compound may include a covalent bond formed by the reaction between the epoxy group of the second compound and the amine group of the third compound. Furthermore, the fourth compound may include a covalent bond formed by the reaction between the epoxy group of the second compound and the carboxyl group of the third compound. That is, the fourth compound may not be a simple mixture of the second compound and the third compound.

[0118] If the content of the fourth compound satisfies the above-described numerical range, a metal oxide on the semiconductor chip **100**, the semiconductor substrate **1000**, and the plurality of first and second conductive pads may be effectively removed, and the formation of the protective pattern may be effectively performed. For example, if the content of the fourth compound is less than 3 wt % a metal oxide may not be removed effectively, the content of the fourth compound is greater than 33 wt % the protective pattern may not be formed effectively.

[0119] The content of the photoinitiator may be 2 wt % to 10 wt % based on 100 wt % of the flux agent. If the content of the photoinitiator is less than 2 wt %, a photo-curing reaction may not effectively occur. If the content of the photoinitiator is greater than 10 wt %, a metal oxide on the semiconductor chip **100** and the semiconductor substrate **1000** may not be effectively removed. The flux agent may further include a catalyst, a solvent, a leveling agent, and metal nanoparticles.

[0120] The rest of the components may be substantially the same as those of the semiconductor package described with reference to FIG. 1 and FIG. 2.

[0121] FIG. 21 to FIG. 25 are views showing a method for manufacturing a semiconductor package according to some embodiments of the inventive concept, and are cross-sectional views corresponding to A-A' of FIG. 1. In order to simplify the description, the same descriptions as those of the method for manufacturing a semiconductor package described with reference to FIG. 1 and FIG. 8 will be omitted.

[0122] Referring to FIG. 21, a flux agent **510P** may be formed on a first surface **100a** of a semiconductor chip **100**. The flux agent **510P** may cover the first surface **100a** of the semiconductor chip **100**, and may cover all of a plurality of second conductive pads **110** and a plurality of connecting bumps **210** on the first surface **100a** of the semiconductor chip **100**. The forming of the flux agent **510P** on the first surface **100a** of the semiconductor chip **100** may include, for example, a spin coating process.

[0123] Referring to FIG. 22, a preliminary protective layer **510L** may be formed. The preliminary

protective layer **510L** may cover the plurality of second conductive pads **110** and the plurality of connecting bumps **210**. The forming of the preliminary protective layer **510L** may include irradiating the flux agent **510P** with light or heat energy, thereby curing the flux agent **510P**.

[0124] By the light energy or heat energy, a curing reaction by a photoinitiator may occur. A curing reaction of a first compound of the flux agent **510P** may occur. Accordingly, the flux agent **510P** may be cured. The light energy may include, for example, light (e.g., ultraviolet light) having a wavelength of 10 nm to 600 nm.

[0125] A temporary tape **120** may be formed on the first surface **100a** of the semiconductor chip **100**. The temporary tape **120** may be formed on the preliminary protective layer **510L**, and may conformally cover the preliminary protective layer **510L**.

[0126] A back-grinding process **630** may be performed on a second surface **100b** of the semiconductor chip **100**. Due to the back-grinding process **630**, a thickness **100D** of the semiconductor chip **100** in a third direction **D 3** may decrease. Accordingly, the degree of integration of the semiconductor package may be improved. Due to the preliminary protective layer **510L**, the first surface **100a** of the semiconductor chip **100**, the plurality of second conductive pads **110**, and the plurality of connecting bumps **210** may be prevented from having an oxide formed thereon, and may be physically and chemically protected.

[0127] Referring to FIG. **23**, a protective tape **125** may be disposed on the second surface **100b** of the semiconductor chip **100**. The protective tape **125** may conformally cover the second surface **100b** of the semiconductor chip **100**. The protective tape **125** may come into contact with the second surface **100b** of the semiconductor chip **100**. Through the protective tape **125**, the semiconductor chip **100** may be protected from a physical impact. The temporary tape **120** on the first surface **100a** of the semiconductor chip **100** may be removed.

[0128] Referring to FIG. **24**, a dicing process of sawing along a sawing line **SL** between the semiconductor chips **100** may be performed. Individually separated semiconductor chips **100** may be formed.

[0129] An additional cleaning process may be further included to remove foreign substances generated by the dicing process. Through the preliminary protective layer **510L**, the first surface **100a** of the semiconductor chip **100**, the plurality of second conductive pads **110**, and the plurality of connecting bumps **210** may be prevented from having an oxide formed thereon during the cleaning process, and may be physically and chemically protected.

[0130] Referring to FIG. **25**, a plurality of first conductive pads **1010** may be formed on an upper surface **1000a** of a semiconductor substrate **1000**. The first surface **100a** of the semiconductor chip **100** may be provided to face the upper surface **1000a** of the semiconductor substrate **1000**. Each of the plurality of second conductive pads **110** and the plurality of connecting bumps **210** may be alignment to correspond to a corresponding first conductive pad **1010**.

[0131] A bonding process **620** may be performed. The bonding process **620** may include applying pressure to the semiconductor chip **100** and the semiconductor substrate **1000** which are aligned. The bonding process **620** may include applying pressure to the semiconductor chip **100** and the semiconductor substrate **1000**, and applying light energy or heat energy to the plurality of preliminary protective patterns **500P**.

[0132] Referring back to FIG. **1** and FIG. **20**, a protective layer **500L** and a plurality of protective patterns **500** may be formed. The forming of the protective layer **500L** and the plurality of protective patterns **500** may include, through above-described bonding process, applying pressure to the preliminary protective layer **510L**, and applying heat or light energy to the preliminary protective layer **510L**. The light energy may include, for example, a laser having a wavelength of 600 nm to 3000 nm.

[0133] Due to the heat energy or light energy in the bonding process, the preliminary protective layer **510L** may have fluidity. A portion of the preliminary protective layer **510L** may be pushed to side surfaces of the plurality of first and second conductive pads **1010** and **110** and the plurality of

connecting bumps **210**. The portion of the preliminary protective layer **510L** may cover side surfaces **1010S**, **110S**, and **500S** of the plurality of first and second conductive pads **1010** and **110** and the plurality of connecting bumps **210**. The portion of the preliminary protective layer **510L** may come into contact with the upper surface **1000a** of the semiconductor substrate **1000**. The portion of the preliminary protective layer **510L** may form the plurality of protective patterns **500**. [0134] Furthermore, due to the heat energy or light energy, a fourth compound in the preliminary protective layer **510L** may be activated. Accordingly, a metal oxide on the plurality of connecting bumps **210** may be removed.

[0135] The rest of the manufacturing method may be substantially the same as the manufacturing method of the semiconductor package described with reference to FIG. **1** and FIG. **8**.

[0136] FIG. **26** is a cross-sectional view according to some embodiments of the inventive concept, and is a cross-sectional view taken along line A-A' of FIG. **1**. In order to simplify the description, the same descriptions as those of the semiconductor package described with reference to FIG. **1** and FIG. **2** will be omitted.

[0137] Referring to FIG. **1** and FIG. **26**, a buffer chip **1200** including an upper surface **1200a** and a lower surface **1200b** which are opposite to each other may be provided. The buffer chip **1200** may include, for example, an interposer or a logic circuit chip.

[0138] A plurality of first conductive pads **1010** may be disposed on an upper surface **1200a** of the buffer chip **1200**. The plurality of first conductive pads may be spaced apart from each other in a first direction **D1** and a second direction. A lower external pad **1210** and a lower external connecting terminal **1220** on the lower external pad **1210** may be disposed on a lower surface **1200b** of the buffer chip **1200**.

[0139] A plurality of semiconductor chips MC may be disposed on the buffer chip **1200**. The plurality of semiconductor chips MC may include, for example, a first semiconductor chip **S1**, a second semiconductor chip **S2**, and a third semiconductor chip **S3**. The first and second semiconductor chips **S1** and **S2** may include, for example, any one of DRAM, NAND flash, SRAM, MRAM, PRAM, and RRAM. The third semiconductor chip **S3** may be, for example, a dummy chip.

[0140] The first semiconductor chip **S1** may be a semiconductor chip closest to the buffer chip **1200**, and the third semiconductor chip **S3** may be a semiconductor chip farthest from the buffer chip **1200**. The second semiconductor chip **S2** may be a semiconductor chip interposed between the first semiconductor chip **S1** and the third semiconductor chip **S3**. The number of the second semiconductor chips **S2** is not limited to what is shown in FIG. **26**. The number of the second semiconductor chips **S2** may be plural.

[0141] The first semiconductor chip **S1** may be connected to the buffer chip **1200** through a plurality of first connecting bumps **211**. More specifically, the plurality of first connecting bumps **211** may be interposed between the plurality of first conductive pads **1010** of the buffer chip **1200** and a plurality of second conductive pads **110** on a first surface **100a** of the first semiconductor chip. Accordingly, the first semiconductor chip **S1** may be electrically connected to the buffer chip **1200**. The plurality of first connecting bumps **211** may be, for example, solder.

[0142] A plurality of first protective patterns **501** may be interposed between the first semiconductor chip **S1** and the buffer chip **1200**. The plurality of first protective patterns **501** may cover the plurality of first connecting bumps **211**. The plurality of first protective patterns **501** may correspond to the plurality of protective patterns **500** described with reference to FIG. **1** and FIG. **2**.

[0143] Each of the first to third semiconductor chips **S1**, **S2**, and **S3** may include substantially the same configuration as each other. Each of the first to third semiconductor chips **S1**, **S2**, and **S3** may include a first surface **100a** and a second surface **100b** which are opposite to each other. Each of the first to third semiconductor chips **S1**, **S2**, and **S3** may include a plurality of second conductive pads **110** on the first surface **100a**, and may include a plurality of third conductive pads **111** on the second surface **100b**. The plurality of third conductive pads **111** on the second surface **100b** of the

third semiconductor chip **S3** may be omitted.

[0144] Each of the first and second semiconductor chips **S1** and **S2** may include a circuit layer **101** on the first surface **100a**, a semiconductor layer **102** on the circuit layer **101**, and a passivation layer **103** on the semiconductor layer **102**. Each of the first and second semiconductor chips **S1** and **S2** may include a through-via **100V** passing through the inside of the first and second semiconductor chips **S1** and **S2**. The third semiconductor chip **S3** may not include a circuit layer, a semiconductor layer, and a through-via. Due to the formation of the dummy chip, defects in a process may be reduced, and a heat management system may be improved.

[0145] The first to third semiconductor chips **S1**, **S2**, and **S3** may be electrically connected to each other through a plurality of second connecting bumps **212**. More specifically, the plurality of second connecting bumps **212** may be disposed between the second surface **100b** of the first semiconductor chip **S1** and the first surface **100a** of the second semiconductor chip **S2**. The plurality of second connecting bumps **212** may be disposed between the second surface **100b** of the second semiconductor chip **S2** and the first surface **100a** of the third semiconductor chip **S3**. The plurality of second connecting bumps **212** may be disposed between the plurality of second conductive pads **110** and the plurality of third conductive pads **111**. Accordingly, the first to third semiconductor chips **S1**, **S2**, and **S3** may be electrically connected to each other. The plurality of second connecting bumps **212** may be, for example, solder.

[0146] A plurality of second protective patterns **502** may be interposed between the first to third semiconductor chips **S1**, **S2**, and **S3**. Each of the plurality of second protective patterns **502** may come into contact with the first surface **100a** and the second surface **100b** of each of the first to third semiconductor chips **S1**, **S2**, and **S3**. The plurality of second protective patterns **502** may cover the plurality of second and third conductive pads **110** and **111**. The plurality of second protective patterns **502** may cover the plurality of second connecting bumps **212**. The plurality of first protective patterns **502** may correspond to the plurality of protective patterns **500** described with reference to FIG. 2 and FIG. 2.

[0147] A molding film **400** may be disposed on the upper surface **1200a** of the buffer chip **1200**. The molding film **400** may cover a plurality of semiconductor chips **MC**, and may fill a space between the plurality of second protective patterns **502**. The molding film **400** may come into contact with the first surface **100a** and the second surface **100b** of each of the first to third semiconductor chips **S1**, **S2**, and **S3**. The molding film **400** may comprise, for example, an epoxy molding compound, a molded underfill.

[0148] The rest of the components may be substantially the same as those of the semiconductor package described with reference to FIG. 1 and FIG. 2.

[0149] FIG. 27 is a plan view of a semiconductor package according to some embodiments of the inventive concept. FIG. 28 is a cross-sectional view of a semiconductor package according to some embodiments of the inventive concept, and is a cross-sectional view taken along A-A' of FIG. 27. In order to simplify the description, the same descriptions as those of the semiconductor package described with reference to FIG. 1 and FIG. 2 will be omitted.

[0150] Referring to FIG. 27 and FIG. 28, a package substrate **1600**, an interposer substrate **1500** on the package substrate **1600**, a first semiconductor substrate **1300** on the interposer substrate **1500**, a graphics processing unit **700** on the interposer substrate **1500**, and a chip stack **CS** on the first semiconductor substrate **1300** may be included.

[0151] The package substrate **1600** may be, for example, any one of a printed circuit board, a semiconductor chip, and a semiconductor package. The package substrate **1600** may include upper pads **1603** on an upper surface thereof. The package substrate **1600** may include lower pads **1610** on a lower surface thereof. The lower pads **1610** may be electrically connected to the upper pads **1603** through internal wirings disposed inside the package substrate **1600**. The package substrate **1600** may include lower bumps **1620** on the lower surface thereof, and each of the lower bumps **1620** may be disposed on a corresponding lower pad among the lower pads **1610**.

[0152] The interposer substrate **1500** may rewire the first semiconductor substrate **1300** and the graphics processing unit **700**. The interposer substrate **1500** may be mounted on the package substrate **1600** in a flip-chip manner.

[0153] The interposer substrate **1500** may include first connecting pads **1520** on a lower surface thereof. The interposer substrate **1500** may include first bumps **1530** on the lower surface thereof. Each of the first bumps **1530** may be electrically connected to a corresponding first connecting pad among the first connecting pads **1520**. A first underfill **1540** may be disposed between the interposer substrate **1500** and the package substrate **1600**. The first underfill **1540** may cover the first bumps **1530**.

[0154] Second connecting pads **1510** may be disposed on an upper surface of the interposer substrate **1500**. The second connecting pads **1510** may be electrically connected to the first connecting pads **1520** through internal wirings disposed inside the interposer substrate **1500**.

[0155] The first semiconductor substrate **1300** may be connected to the interposer substrate **1500** through external terminals **1310**. The external terminals **1310** may be electrically connected to the second connecting pads **1510** of the interposer substrate **1500**. The first semiconductor substrate **1300** may correspond to the buffer chip **1200** described with reference to FIG. 26. The chip stack CS may be disposed on the first semiconductor substrate **1300**. The chip stack CS may correspond to the plurality of semiconductor chips MC described with reference to FIG. 26. A molding film **400** may cover the first semiconductor substrate **1300** and the chip stack CS. The molding film **400** may correspond to the molding film described with reference to FIG. 26.

[0156] The graphics processing unit **700** may be disposed on the interposer substrate **1500**. The graphics processing unit **700** may be spaced apart from the first semiconductor substrate **1300** in the first direction D1. A thickness of the graphics processing unit **700** may be greater than a thickness of each of semiconductor chips of the chip stack CS. The graphics processing unit GPU **700** may be, for example, a logic chip. Pads **710** and bumps **720** on the pads **710** may be disposed on a lower surface of the graphics processing unit **700**. The graphics processing unit **700** may be electrically connected to the interposer substrate **1500** through the bumps **720**. Lower protective patterns **730** may cover the pads **710** and the bumps **720**. The configuration of the lower protective pattern **730** may correspond to that of the plurality of protective patterns described with reference to FIG. 1 and FIG. 2.

[0157] An outer molding film **450** may be disposed on the package substrate **1600**. The outer molding film **450** may cover the interposer substrate **1500**, the first semiconductor substrate **1300**, the chip stack CS, and the graphics processing unit **700**. The outer molding film may comprise, for example, an epoxy molding compound, a molded underfill.

[0158] The rest of the components may be substantially the same as those of the semiconductor package described with reference to FIG. 1 and FIG. 2.

[0159] FIG. 29 is a plan view of a semiconductor package according to some embodiments of the inventive concept. FIG. 30 is a cross-sectional view of a semiconductor package according to some embodiments of the inventive concept, and is a cross-sectional view taken along A-A' of FIG. 29. In order to simplify the description, the same descriptions as those of the semiconductor package described with reference to FIG. 1 and FIG. 2 will be omitted.

[0160] Referring to FIG. 29 and FIG. 30, a plurality of first conductive pads **1010** may be disposed on an upper surface **1000a** of a semiconductor substrate **1000**. A first semiconductor die **151** and a second semiconductor die **152** may be disposed on the upper surface **1000a** of the semiconductor substrate **1000**.

[0161] The first semiconductor die **151** may include, for example, a logic chip. The second semiconductor die **152** may include, for example, a memory chip, and may include, for example, a volatile memory semiconductor chip such as dynamic random access memory (DRAM) or static random access memory (SRAM). The second semiconductor die **152** may include, for example, a non-volatile memory semiconductor chip such as a flash memory, a phase-change random access

memory (PRAM), a magnetic random access memory (MRAM), a ferroelectric random access memory (FeRAM), or a resistive random access memory (RRAM). However, the inventive concept is not limited thereto, and the first semiconductor die **151** may include a memory chip.

[0162] The first semiconductor die **151** may include a third surface **151a** and a fourth surface **151b** which are opposite to each other. The third surface **151a** of the first semiconductor die **151** may face the upper surface **1000a** of the semiconductor substrate **1000**. A plurality of second conductive pads **110** may be disposed on the third surface **151a** of the first semiconductor die **151**. The plurality of second conductive pads **110** may be disposed to correspond to the plurality of first conductive pads **1010**.

[0163] The second semiconductor die **152** may include a fifth surface **152a** and a sixth surface **152b** which are opposite to each other. The fifth surface **152a** of the second semiconductor die **152** may face the upper surface **1000a** of the semiconductor substrate **1000**. A plurality of third conductive pads **121** may be disposed on the fifth surface **152a** of the second semiconductor die **152**. The plurality of third conductive pads **121** may be disposed to correspond to the plurality of first conductive pads **1010**.

[0164] A plurality of connecting bumps **210** may be disposed between the plurality of first conductive pads **1010** and the plurality of second conductive pads **110** and between the plurality of first conductive pads **1010** and the plurality of third conductive pads **121**. Accordingly, the first semiconductor die **151** and the semiconductor substrate **1000** may be electrically connected to each other, and the second semiconductor die **152** and the semiconductor substrate **1000** may be electrically connected to each other. The plurality of connecting bumps **210** may be solder.

[0165] A plurality of protective patterns **500** may be disposed between the first semiconductor die **151** and the semiconductor substrate **1000** and between the second semiconductor die **152** and the semiconductor substrate **1000**. Each of the plurality of protective patterns **500** may come into contact with the third surface **151a** of the first semiconductor die **151**, and may come into contact with the fifth surface **152a** of the second semiconductor die **152**. Each of the plurality of protective patterns **500** may come into contact with the upper surface **1000a** of the semiconductor substrate **1000**.

[0166] Each of the plurality of protective patterns **500** may cover corresponding first and second conductive pads among the plurality of first and second conductive pads **1010** and **110**. Each of the plurality of protective patterns **500** may cover corresponding first and third conductive pads among the plurality of first and third conductive pads **1010** and **121**. Each of the plurality of protective patterns **500** may cover a corresponding connecting bump among the plurality of connecting bumps **210**. The plurality of protective patterns **500** may correspond to the plurality of protective patterns **500** described with reference to FIG. 1 and FIG. 2.

[0167] A molding film **400** may be disposed on the upper surface **1000a** of the semiconductor substrate **1000**. The molding film **400** may cover the first and second dies **151** and **152**, and may fill a space between the plurality of protective patterns **500**. The molding film **400** may come into contact with the upper surface **1000a** of the semiconductor substrate **1000**, and may come into contact with the third surface **151a** of the first semiconductor die **151** and the fifth surface **152a** of the second semiconductor die **152**.

[0168] The molding film **400** may come into contact with a side surface **500S** of the plurality of protective patterns **500**. The molding film **400** may include, for example, an epoxy molding compound. The molding film **400** may protect the semiconductor package from impurities, and external impacts.

[0169] The rest of the components may be substantially the same as those of the semiconductor package described with reference to FIG. 1 and FIG. 2.

[0170] Hereinafter, the inventive concept will be described with reference to the following example.

Example

[0171] 3,4-Epoxy cyclohexylmethyl 3,4-epoxy cyclohexanecarboxylate was prepared as a first compound. 3-Glycidoxypropyltrimethoxysilane was prepared as a second compound. L-Lysine was prepared as a third compound. 12.5 g of the second compound and 5.2 g of the third compound were reacted in flask at 181° C. for 5 hours to form the fourth compound. triarylsulfonium hexafluorophosphate salt was prepared as a photoinitiator. 1,5,7-triazabicyclo[4.4.0]dec-5-ene was prepared as a catalyst. Tin was prepared from metal nanoparticles, and the average particle diameter of the metal nanoparticles was 0.1 μm. The first compound, the fourth compound, the photoinitiator, the catalyst, and the metal nanoparticles were mixed in acetone to form a flux agent. In this case, the content of the first compound was 81 wt % based on 100 wt % of the flux agent solid, and the content of the fourth compound was 13 wt % based on 100 wt % of the flux agent solid. The content of the photoinitiator was 4 wt % based on 100 wt % of flux agent solid.

Evaluation Example: Scanning Electron Microscope (SEM) Photograph

[0172] FIG. 31A is a result of photographing a semiconductor chip using a scanning electron microscope. FIG. 31B is a result of photographing a semiconductor substrate using a scanning electron microscope.

[0173] FIG. 32 is a result of photographing a protective pattern according to the embodiment formed on the semiconductor chip according to FIG. 31A. The flux agent according to the embodiment was applied on the semiconductor chip and then cured by being applied with light energy to form a protective pattern. The light energy was ultraviolet light having a wavelength of 365 nm. Thereafter, an image was photographed through a scanning electron microscope.

[0174] FIG. 33 is result of photographing a cross-section of a junction after performing a bonding process on the semiconductor chip having a protection pattern formed in FIG. 32 to the semiconductor substrate of FIG. 31B. The bonding process was performed by applying a pressure of about 1.5N to the semiconductor chip and the semiconductor substrate while applying heat by irradiating the semiconductor chip with a laser having 980 nm wavelength.

[0175] Referring to FIG. 32 and FIG. 33, it can be seen that the flux agent was cured, so that the protective pattern was formed. Furthermore, it can be confirmed that the protective pattern was formed in a solid state.

Evaluation Example: Evaluation of Electrical Resistance

[0176] The electrical resistance of the solder bump formed between the semiconductor chip and the semiconductor substrate bonded by the bonding process of FIG. 32 and FIG. 33 was measured three times.

TABLE-US-00001 TABLE 1 Electrical resistance (mΩ) 1 time.sup. 4.7 2 times 4.7 3 times 4.79
Average value 4.73

[0177] Referring to Table 1, it can be confirmed that the electrical resistance of the solder bump is as low as about 4.73 mΩ. Therefore, it can be confirmed that the flux agent according to the embodiment and the protective pattern, which is a cured product thereof, may implement both the flux function and the underfill function.

[0178] According to embodiments of the inventive concept, a protective pattern may be a cured product of a flux agent. Accordingly, the protective pattern may perform both a flux function of removing a metal oxide from a semiconductor chip, a semiconductor package, a conductive pad, and connecting bumps, and an underfill function of protecting the same.

[0179] Furthermore, the protective pattern may be in a solid state at room temperature as a cured product of the flux agent. A bonding portion may be physically and chemically protected through the protective pattern before a bonding process of a semiconductor substrate and a semiconductor chip. In addition, in the bonding process, a risk in which the protective pattern flows may be eliminated. Accordingly, the difficulty of a manufacturing process of a semiconductor package may be reduced.

[0180] Furthermore, after the formation of the protective pattern, the semiconductor chip and the semiconductor substrate may be stored for a long period of time. Accordingly, application fixation

and the bonding process of the flux agent may be spatially and temporally dualized. Accordingly, a method for manufacturing various semiconductor packages may be applied.

[0181] Although the embodiments of the inventive concept have been described with reference to the accompanying drawings, those skilled in the art will appreciate that the present invention can be variously modified and changed without departing from the spirit and scope of the present invention as set forth in the following patent claims. In addition, the embodiments disclosed herein are not intended to limit the technical spirit of the present invention, and all technical concepts falling within the scope of the following claims and equivalents thereof are to be construed as being included in the scope of the inventive concept.

Claims

1. A flux agent comprising: a first compound comprising an ester group and an epoxy group; a fourth compound which is a chemical reactant of a second compound comprising an epoxy group and a third compound comprising an amine group and a carboxyl group; and a photoinitiator, wherein: the content of the first compound is 60 wt % to 95 wt % based on 100 wt % of the total agent; the content of the fourth compound is 3 wt % to 33 wt % based on 100 wt % of the total agent; and the content of the photoinitiator is 2 wt % to 10 wt % based on 100 wt % of the total agent.
2. The flux agent of claim 1, wherein the first compound and the second compound each comprise at least one of glycidyl epoxy or cyclohexene oxide.
3. The flux agent of claim 1, wherein the first compound comprises one or more selected from the group consisting of Methyl 3-(7-oxabicyclo[4.1.0]heptan-3-yl)-7-oxabicyclo[4.1.0]heptane-3-carboxylate, (5-Methyl-7-oxabicyclo[4.1.0]heptan-3-yl)methyl 5-methyl-7-oxabicyclo[4.1.0]heptane-3-carboxylate, [2,2-Dimethyl-3-(6-methyl-7-oxabicyclo[4.1.0]heptane-3-carbonyl)oxypropyl] 6-methyl-7-oxabicyclo[4.1.0]heptane-3-carboxylate, (1-Methyl-7-oxabicyclo[4.1.0]heptan-3-yl)methyl 1-methyl-7-oxabicyclo[4.1.0]heptane-3-carboxylate, 7-Oxabicyclo[4.1.0]heptan-2-ylmethyl 7-oxabicyclo[4.1.0]heptane-3-carboxylate, 6-(7-Oxabicyclo[4.1.0]heptan-1-ylmethyl)-7-oxabicyclo[4.1.0]heptane-3-carboxylate, (2-Methyl-7-oxabicyclo[4.1.0]heptan-3-yl)methyl 2-methyl-7-oxabicyclo[4.1.0]heptane-3-carboxylate, 2-(7-Oxabicyclo[4.1.0]heptan-3-yl) ethyl 7-oxabicyclo[4.1.0]heptane-3-carboxylate, 2-[2-(7-Oxabicyclo[4.1.0]heptane-3-carbonyloxy)ethoxy]ethyl 7-oxabicyclo[4.1.0]heptane-3-carboxylate, Bis(7-oxabicyclo[4.1.0]heptan-3-ylmethyl) heptanedioate, 3,4-Epoxy cyclohexylmethyl 3,4-epoxycyclohexanecarboxylate, Bis(3,4-epoxycyclohexylmethyl) adipate, Chissonox 201, 3-[2-(3-Carboxylato-7-oxabicyclo[4.1.0]heptan-3-yl)ethyl]-7-oxabicyclo[4.1.0]heptane-3-carboxylate, 2-(7-Oxabicyclo[4.1.0]heptane-3-carbonyloxy) ethyl 7-oxabicyclo[4.1.0]heptane-3-carboxylate, 3,4-Epoxy cyclohexyl-3,4-epoxycyclo-hexane carboxylate, 5-Methyl-3-[(5-methyl-7-oxabicyclo[4.1.0]heptan-3-yl)methyl]-7-oxabicyclo[4.1.0]heptane-3-carboxylate, 1,4-Cyclohexanedimethanol bis(3,4-epoxycyclohexanecarboxylate), Cyclohexylmethyl-3,4-epoxycyclohexanecarboxylate, Glycidyl methacrylate, Ethyl 2,3-epoxybutyrate, Methyl 3,4-epoxybutyrate, Ethyl oxirane-2-carboxylate, Glycidyl butyrate, (2-Methyloxiranyl)methyl methacrylate, 9,10-Epoxy stearate, Diglycidyl isophthalate, oxiran-2-ylmethyl benzoate, Tris(oxiranylmethyl)benzene-1,3,5-tricarboxylate, Bis(2,3-epoxypropyl) terephthalate, Bis(oxiran-2-ylmethyl) 5-carbonochloridoylbenzene-1,3-dicarboxylate, and a combination thereof.
4. The flux agent of claim 1, wherein the third compound comprises one or more selected from the group consisting of alpha (α)-amino acid, beta (β)-amino acid, gamma (γ)-amino acid, delta (δ)-amino acid, anthranilic acid, 3-aminobenzoic acid, para-aminobenzoic acid, and a combination thereof.
5. The flux agent of claim 1, wherein the photoinitiator comprises one or more selected from the group consisting of 3-methyl-2-butenyltetramethylenesulfonium hexafluoroantimonate salt,

ytterbium trifluoromethanesulfonate salt, samarium trifluoromethanesulfonate salt, erbium trifluoromethanesulfonate salt, triarylsulfonium hexafluoroantimonate salt, triarylsulfonium hexafluorophosphate salt, lanthanum trifluoromethanesulfonate salt, tetrabutylphosphonium methanesulfonate salt, ethyltriphenylphosphonium bromide salt, diphenyldiodonium hexafluoroantimonate salt, diphenyldiodonium hexafluorophosphate salt, dithoriliodonium hexafluorophosphate salt, 9-(4-hydroxyethoxyphenyl)cyanthrene hexafluorophosphate salt, 1-(3-methylbut-2-enyl) tetrahydro-1H-thiophenium hexafluoroantimonate salt, 2-(9-oxoxanthen-2-yl) propionic acid 1,5,7-triazabicyclo[4.4.0]dec-5-ene salt, and a combination thereof.

6. The flux agent of claim 1, further comprising metal nanoparticles, wherein the metal nanoparticles contain one or more selected from the group consisting of tin, copper, silver, bismuth, indium, lead, cadmium, antimony, gallium, arsenic, germanium, zinc, aluminum, gold, silicon, nickel, phosphorus, and an alloy thereof.

7. The flux agent of claim 6, wherein the metal nanoparticles have an average particle diameter of 0.01 μm to 1 μm .

8. The flux agent of claim 1, further comprising a catalyst, wherein the catalyst includes one or more selected from the group consisting of 1,5,7-triazabicyclo[4.4.0]dec-5-ene, 2-methylimidazole, 1-methylimidazole, zinc acetate, triphenylphosphine, dibutyl phosphate, and a combination thereof.

9. A semiconductor package comprising: a semiconductor substrate comprising an upper surface and a lower surface which are opposite to each other; a semiconductor chip including a first surface and a second surface which are opposite to each other, and disposed on the upper surface of the semiconductor substrate, wherein the first surface faces the upper surface of the semiconductor substrate; and a plurality of protective patterns interposed between the semiconductor substrate and the semiconductor chip, wherein each of the plurality of protective patterns is a cured product of a flux agent, wherein the flux agent comprises: a first compound comprising an ester group and an epoxy group; a fourth compound which is a chemical reactant of a second compound comprising an epoxy group and a third compound comprising an amine group and a carboxyl group; and a photoinitiator, wherein: the content of the first compound is 60 wt % to 95 wt % based on 100 wt % of the flux agent; the content of the fourth compound is 3 wt % to 33 wt % based on 100 wt % of the flux agent; and the content of the photoinitiator is 2 wt % to 10 wt % based on 100 wt % of the flux agent.

10. The semiconductor package of claim 9, further comprising: a plurality of first conductive pads on the upper surface of the semiconductor substrate; a plurality of second conductive pads on the first surface of the semiconductor chip; and a plurality of connecting bumps interposed between the plurality of first conductive pads and the plurality of second conductive pads, wherein each of the plurality of protective patterns covers a side surface of a corresponding connecting bump among the plurality of connecting bumps.

11. The semiconductor package of claim 9, wherein each of the plurality of protective patterns is in contact with the upper surface of the semiconductor substrate.

12. The semiconductor package of claim 9, wherein each of the plurality of protective patterns covers a side surface of a corresponding first conductive pad among the plurality of first conductive pads and a side surface of a corresponding second conductive pad among the plurality of second conductive pads.

13. The semiconductor package of claim 9, wherein each of the plurality of protective patterns is in contact with the first surface of the semiconductor chip.

14. The semiconductor package of claim 9, further comprising a molding film disposed on the semiconductor substrate, and covering the semiconductor chip and the plurality of protective patterns.

15. The semiconductor package of claim 14, wherein the plurality of protective patterns are spaced apart from each other in a first direction, and the molding film fills a space between the plurality of protective patterns, wherein the first direction is a direction parallel to the upper surface of the

semiconductor substrate.

16. The semiconductor package of claim 9, wherein a width in a first direction of each of the plurality of protective patterns is smaller than a width in the first direction of the semiconductor chip, wherein the first direction is a direction parallel to the upper surface of the semiconductor substrate.

17. The semiconductor package of claim 9, wherein a width in a first direction of each of the plurality of protective patterns decreases in a third direction farther from the upper surface of the semiconductor substrate, wherein the first direction is a direction parallel to the upper surface of the semiconductor substrate, wherein the third direction is a direction perpendicular to the upper surface of the semiconductor substrate.

18. The semiconductor package of claim 9, wherein the first compound and the second compound each comprise at least one of glycidyl epoxy and cyclohexene oxide.

19. A semiconductor package comprising: a semiconductor substrate including an upper surface and a lower surface which are opposite to each other; a plurality of first conductive pads on the upper surface of the semiconductor substrate; a semiconductor chip comprising a first surface and a second surface which are opposite to each other, and disposed on the upper surface of the semiconductor substrate, wherein the first surface faces the upper surface of the semiconductor substrate; a plurality of second conductive pads on the first surface of the semiconductor chip; a plurality of connecting bumps interposed between the plurality of first conductive pads and the plurality of second conductive pads; and a plurality of protective patterns covering the plurality of connecting bumps, wherein a width in a first direction of each of the plurality of protective patterns decreases in a third direction farther from the upper surface of the semiconductor substrate, wherein the first direction is a direction parallel to the upper surface of the semiconductor substrate, wherein the third direction is a direction perpendicular to the upper surface of the semiconductor substrate.

20. The semiconductor package of claim 19, wherein each of the plurality of protective patterns is a cured product of a flux agent, wherein the flux agent comprises: a first compound including an ester bond and an epoxy group; a fourth compound which is a chemical reactant of a second compound including an epoxy group and a third compound including an amine group and a carboxyl group; and a photoinitiator.
